

ORIGINAL RESEARCH

A deep learning and ensemble learning based architecture for metro passenger flow forecast

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Abstract

Accurate short-term forecast of metro outbound passenger flow is of great significance for real-time traffic control and guidance. A good forecast method should have high accuracy, timeliness and practicality. Based on deep learning and ensemble learning technology, this study proposes an end-to-end forecast hybrid architecture for metro outbound passenger flow that integrates multiple passenger flow features. The architecture innovatively integrates bagging ensemble learning strategy and transfer learning with deep learning, and includes multiple extensible passenger flow feature processing components. In addition, this study presents a new coding method to incorporate the operating characteristics of the metro into the forecasting architecture. Use the automatic fare collection (AFC) data of Chengdu Metro Tianfu Square Station for training and verify the forecast architecture on workdays, weekends and holidays. The results reveal that compared with other widely used passenger flow forecast models, the architecture proposed in this study has achieved the highest forecast accuracy in the above-mentioned different time types. Furthermore, the fusion of transfer learning improves the accuracy of forecast model while significantly speeding up the convergence of training, increasing its timeliness and practicability.

1 | INTRODUCTION

As an efficient, large-volume and sustainable transportation mode, the metro has become an important part of urban public transportation. Accurate forecast of the short-term outbound passenger flow of the metro can control and guide the passenger flow to better connect with other transportation modes, and contribute to better arrangement of shuttles, especially in large transit hubs, thereby reducing urban traffic congestion [1]. With the rapid development of urban intelligent transportation systems, real-time traffic control and guidance has become the research hotspots in transportation [2]. The key factor to achieve better traffic control and guidance for the metro system is the accuracy and reliability of passenger flow forecast. Short-term passenger flow forecast is the core of the intelligent transportation system [3–5]. Compared with long-term passenger flow, short-term passenger flow reveals less regularity because of its short time span. Therefore, high-precision and high-efficiency short-term passenger flow forecast is more challenging in comparison [1, 4, 6].

The metro short-term passenger flow forecast belongs to the field of traffic flow forecast. There have been a large number of related studies, which are mainly divided into three categories: The first category is various classical linear models based on statistical theory, including the autoregressive integrated moving average (ARIMA)-based models [7–10], exponential smoothing (ES)-based models [11] and Kalman filter-based models [12, 13]. Early study on traffic flow forecast is mostly based on such methods.

Because traffic flow data has the characteristics of time-variance, randomness and strong non-linearity, the first category of the model is difficult to capture the non-linearity and uncertainty of traffic flow. Therefore data-driven machine learning models are now widely used to deal with the above problems. The second category is based on traditional data-driven machine learning methods, such as random forest, support vector machines (SVM). Hou et al. [14] established a random forest, regression tree and non-parametric regression model for urban work zones to make long-term and short-term traffic flow forecast. The results reveal that the random forest model has

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the best performance among the above models. Liu and Yao [15] proposed a least-square support vector machine (LSSVM) that uses an improved particle-swarm optimization algorithm (IPSO) for parameter optimization to forecast metro passenger flow during holidays. Cong et al. [16] proposed an LSSVM forecast model, the parameters of which are determined by the fruit fly optimization algorithm (FOA). Experiments show that its performance is better than LSSVM combined with particle swarm optimization algorithm and radial basis function (RBF) neural network. Zhu et al. [17] utilized linear conditional Gaussian (LCG) and Bayesian networks (BN) to consider both spatial and temporal dimensions of traffic as well as speed information for short-term traffic flow forecast. Sun et al. [18] proposed a hybrid passenger flow forecast model that combines SVM and wavelet. The model regards the passenger flow data as a signal sequence and decomposes it into different frequency sequences, and uses SVM to forecast the sequence. Finally, the diverse forecasted sequences are reconstructed by wavelet. Compared with the first category of model, the second category of model has a certain degree of accuracy improvement due to the increased ability to capture the non-linear characteristics of traffic flow.

However, a large number of current studies have revealed that the forecast performance of deep learning models is generally better than traditional machine learning models [1, 4, 19]. The third category of traffic flow forecast model is based on the deep learning theory. With the continuous development of artificial intelligence technology, various deep learning methods have been studied and successfully applied in the fields of image processing, natural language processing and time series forecast. Deep learning methods can capture various complex non-linear relationships and potential data features in a large amount of historical data, so they are very suitable for traffic flow prediction. Zhao et al. [4] utilized long short-term memory network (LSTM) to forecast short-term traffic flow and compared its performance with SVM, RBF, ARIMA and SAE models. Experiments show that the LSTM model performs best. Yu et al. [20] considered the spatial characteristics of traffic flow and combined convolutional neural networks and LSTM networks to construct a spatiotemporal recurrent convolutional network (SRCN) for traffic flow prediction. Liu et al. [19] proposed an end-to-end deep learning architecture, which is highly flexible and extendable. The architecture comprehensively considers a variety of passenger flow features to forecast the metro outbound passenger flow. Yang et al. [1] proposed an improved spatiotemporal long short-term memory model (Sp-LSTM) to forecast short-term outbound passenger flow at metro stations. Sp-LSTM considers the correlation between the passenger flow of adjacent stations and proposes a method to quantify the correlation of passenger flow between stations. Liu and Chen [21] utilized stacked autoencoder (SAE) to extract historical passenger flow data characteristics, and established an SAE-DNN (Deep neural network) hybrid model to forecast the passenger flow of bus rapid transit stations. Wei and Chen [22] combined empirical mode decomposition (EMD) and multilayer perceptron (MLP), and utilized EMD to decompose and extract passenger flow characteristics, and then input them into the MLP model to forecast. Hao et al. [23] utilized

stacked Bi-directional LSTM (Bi-LSTM) as the encoder network and LSTM as the decoder network to construct a seq2seq forecasting architecture. This architecture introduces the attention mechanism and obtains good forecast performance. Wu et al. [24] fully considered the time characteristics and spatial characteristics of passenger flow based on DNN. And they introduced the attention-based model to capture the important characteristics of the long-term passenger flow data.

The above-mentioned traffic flow forecast studies based on data-driven traditional machine learning and deep learning have still the following three shortcomings. First, the study object is only the passenger flow on workdays or holidays, which limits the application scenarios of the model. Although some studies consider the passenger flow of different time types such as workdays, weekends and holidays, they did not specifically distinguish the different forecast time types when evaluating the performance of the model. Therefore, there is no guarantee that the model has the good forecast accuracy on workdays, weekends and holidays. Second, many forecast models that include traffic flow features are not end-to-end models. The end-to-end model does not need to be trained in components or stages during the training process, but directly optimizes the final prediction target of the model. The training process does not require human intervention or other additional information. Many forecast models are composed of multiple individual components. The training of these components is often performed individually, with separate objective functions. The objective functions might not always be consistent with the overall performance indicator of the system. Thus, the resulting performance of the trained network is not optimized. Third, the metro passenger flow forecast model does not take into account the characteristics of metro operation. Constructing a complete passenger flow feature space can significantly improve the forecast accuracy of the model. The existing studies rarely consider the impact of metro operational properties on outbound passenger flow forecast. As a kind of urban rail transit system, the operation mode of the metro is significantly different from other urban transportation systems such as buses, private cars and taxis. Therefore, when constructing a metro outbound passenger flow forecast model, we cannot simply imitate the traffic flow forecast ideas of other modes of transportation.

Due to the complexity of traffic flow, it is difficult for a single forecast method to obtain high-precision forecast results [9, 10, 25], and a hybrid forecast model that combines multiple methods can usually get better performance. Therefore, in view of the shortcomings of the existing literature, this study proposes an end-to-end forecast hybrid architecture that integrates multiple characteristics of metro outbound passenger flow. This architecture innovatively combines the Bootstrap aggregating (bagging) [26] and transfer learning [27] with deep learning to improve forecast performance and reduce training time.

Ensemble learning is one of the best and most influential machine learning methods [28]. It is considered the state-of-the-art solution for many machine learning challenges [28, 29]. Ensemble learning is the fusion of multiple forecast models, and its combination aims to improve the performance of a single model. Due to this combination of different forecasts,

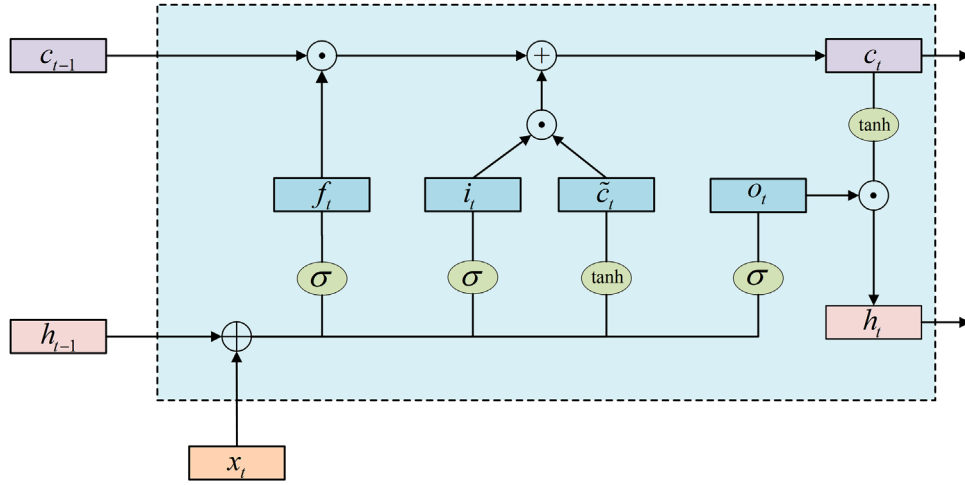


FIGURE 1 Illustration of the hidden layer unit of LSTM model. LSTM, long short-term memory.

ensembles have resulted in very powerful forecast techniques with great generalization without neglecting more local or specific knowledge. Bootstrap Aggregating is an important ensemble learning method, which has been widely used in time series forecast study and other fields [30–34].

Transfer learning is an important and efficient machine learning strategy. Yosinski et al. [27] demonstrated the transferability of features from DNNs and found that neural networks initialized with transferred features exhibit a better generalized performance. Transfer learning can not only simplify manual feature engineering, but also improve the generalization of new models and accelerate model convergence [35]. A large number of studies have revealed that deep learning models based on transfer learning have higher prediction accuracy and greatly reduce model training time [35–37]. This study applies the forecast architecture to different time types such as workdays, weekends and holidays, and specifically analyzes its forecast abilities in different time types. Section 2 summarized some of the existing methods and they are also the basis of our proposed hybrid forecast architecture.

2 | METHODOLOGY

This study proposes an end-to-end forecast hybrid architecture that integrates multiple characteristics of metro outbound passenger flow. This architecture innovatively combines the Bootstrap aggregating (Bagging) and transfer learning with deep learning to improve forecast performance and reduce training time. In Section 2, we introduce the underlying methods used in this architecture, including LSTM, Bi-LSTM, Bagging and transfer learning.

2.1 | LSTM and Bi-LSTM

In order to overcome the vanishing gradient problem, Hochreiter and Schmidhuber [38] proposed an LSTM network and

introduced a gating mechanism. As illustrated in Figure 1, the architecture of LSTM consists of memory cells as basic units. It adds input gate i_t , forget gate f_t , output gate o_t and cell state c_t . The gates can optionally retain the information. These simple changes enable the cell to store and read long-term contextual information, thus solving the vanishing gradient problem. The c_t is used for linear cyclic information transmission, while non-linearly outputting information to the hidden output h_t . And $\tilde{c}_t$ is the candidate state calculated by the non-linear function, the calculation process is as follows:

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (1)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (2)$$

$$h_t = o_t \odot \tanh(c_t) \quad (3)$$

where x_t is the input at time t , h_{t-1} is the hidden state at time $t-1$, c_{t-1} is the cell state at time $t-1$. The operator ‘ $\odot$ ’ is a Hadamard product. Other calculation processes are as follows:

Input gate:

$$i_t = \sigma(W_i [b_{t-1}, x_t] + b_i) \quad (4)$$

Forget gate:

$$f_t = \sigma(W_f [b_{t-1}, x_t] + b_f) \quad (5)$$

Output gate:

$$o_t = \sigma(W_o [b_{t-1}, x_t] + b_o) \quad (6)$$

The σ is logistic function, its output interval is $(0, 1)$, where W_i, W_f and W_o are, respectively, the weight matrices of the input gate, the forget gate and the output gate while b_i, b_f, b_o are their respective bias vectors. The input gate i_t controls the current candidate state how much information needs to be saved. The

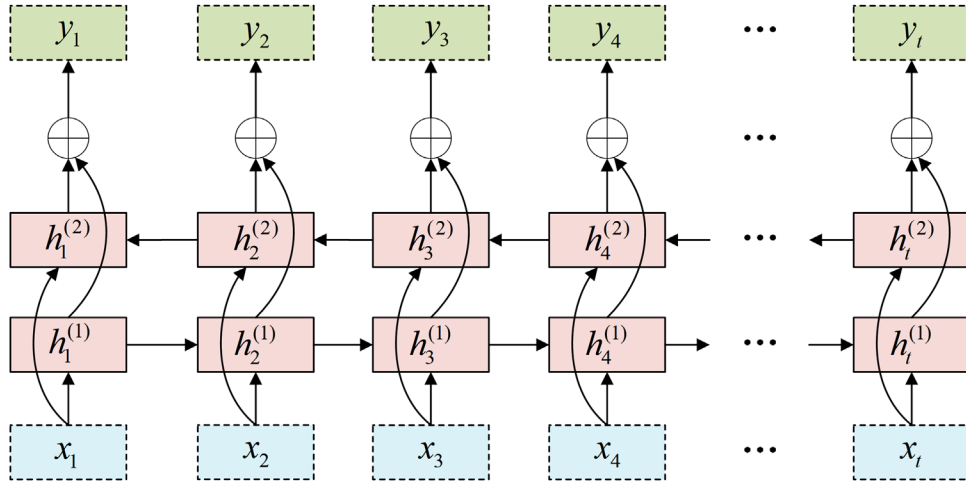


FIGURE 2 Bi-LSTM expanded in chronological order. Bi-LSTM, bi-directional LSTM.

forget gate f_t controls the cell state at time $t - 1$ how much information needs to be forgotten. The output gate o_t control how much information of the cell state c_t needs to be output to the hidden state h_t .

Bi-directional LSTM [39] consists of two layers of LSTM. The input of the two layers is the same, but the transmission direction of data information is opposite. Suppose that the first layer is expanded in a chronological order, and the second layer is expanded in a reverse chronological order, the calculation processes are as follows:

$$h_t^{(1)} = o_t^{(1)} \times \tanh(c_t^{(1)}) \quad (7)$$

$$h_t^{(2)} = o_t^{(2)} \times \tanh(c_t^{(2)}) \quad (8)$$

$$h_t = h_t^{(1)} \oplus h_t^{(2)} \quad (9)$$

where $h_t^{(1)}$ and $h_t^{(2)}$ are the hidden states of the layers at time t , $c_t^{(1)}$ and $c_t^{(2)}$ are the cell states of layers at time t . The operator ' $\oplus$ ' is a vector concatenate. Figure 2 shows Bi-LSTM expanded in a chronological order.

2.2 | Bootstrap aggregating

The bagging algorithm process is as follows: supposing that $D = \{(x_i, y_i) | i = 1, 2, \dots, n\}$ is the initial training set, m samples are randomly selected from the initial training set, and the probability of each sample being selected each time is $1/n$. Build a new data set D_i using the sampled samples. Each sample after being selected is still placed back into the initial training set while constructing the new data set D_i . Repeat sampling K times to get K new data sets $D_1, D_2, \dots, D_K$. Training K forecast models on the above new data set, and finally fuse the K forecast results. SUN [32] and Nikolaos Kourentzes [40] found in their study that taking the median of all forecast results is more accurate

than the mean, so this model uses the median of all forecasts as the fusion result.

2.3 | Transfer learning

Transfer learning is to reuse the knowledge learned in an existing machine learning task in another new similar machine learning task to improve the accuracy and convergence speed of the new task [36]. It is defined as follows: given a source domain D_S and learning task T_S , a target domain D_T and learning task T_T , transfer learning aims to help improve the learning of the target predictive function $f_T(\cdot)$ in D_T using the knowledge in D_S and T_S , where $D_S \neq D_T$, or $T_S \neq T_T$. Figure 3 shows the difference between the traditional machine learning process and the transfer learning process.

3 | METRO PASSENGER FLOW FORECAST HYBRID ARCHITECTURE

Within a fixed interval (e.g. 10-min interval), the total outbound passenger flow could be calculated based on the metro automatic fare collection (AFC) data. We use x_t to denote the outbound passenger flow in the t th time interval. Forecasting outbound passenger flow at fixed time intervals is a typical time series forecast problem, and there is time dependence between consecutive time intervals of passenger flow. Thus, the problem addressed in this paper is to use the historical passenger flow data $\{x_{t-3}, x_{t-2}, x_{t-1}\}$ to forecast x_t . In practical applications, the expectation of obtaining highly accurate forecast results is limited by a number of practical factors, which in this study mainly include the following: first, the experimental data are insufficient. Only some of the metro stations have complete data during the experimental time period. Second, only metro passenger data can be used for forecasting, and other data closely related to passenger flow, such as train diagram data and weather data, are missing. Third, only 1 month of passenger

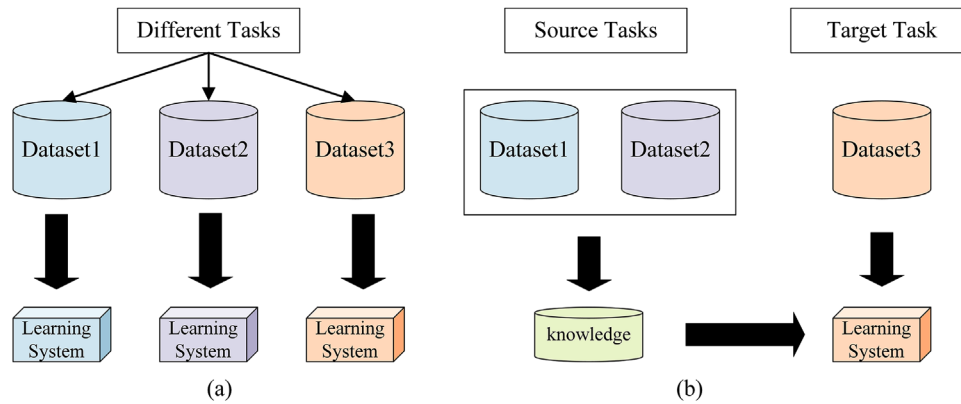


FIGURE 3 Difference between the traditional machine learning process and the transfer learning process

flow data is available, and historical metro passenger flow data for longer time ranges is not available.

We hope to improve the forecast accuracy based on the existing deep learning passenger flow forecast methods, combined with the characteristics of metro passenger flow and usage scenarios. The existing metro passenger flow forecast methods mainly face several challenges: first, how to capture the time-variance, randomness and strong non-linearity of traffic flow more accurately. Second, since the size of the model training set is limited by the data sources and computing equipment, it is necessary to construct the best possible passenger flow feature space, including temporal dependencies, external factors and operational characteristics to provide a preliminary description of passenger flow data and then to better explore the hidden information of passenger flow data and improve the forecast accuracy of the model. Third, how should the passenger flow features be processed and integrated into the training process of neural network for information mining. How to build a forecast model to deal with the above challenges is the critical to improve the accuracy of metro passenger flow forecast. Therefore, this study builds an end-to-end metro passenger flow forecast architecture. The architecture consists of multiple base predictors generated by the bagging method and incorporates transfer learning in the training process to enhance each base predictor's ability to learn from time-variance, randomness and strong non-linearity of passenger flow data. The end-to-end purpose is to integrate various extensible components to improve the forecast ability of the architecture and increase its practicability. The above-mentioned extensible components correspond to each dimension of the passenger flow feature space established by the model, including temporal dependencies, external factors and metro operational properties, respectively. And the neural network architecture could be layer-specific, that is, the different layer can take different neural network model. When designing a deep learning architecture with multi-source input data, an individual module containing multiple layers of the network is designed for each type of data while the modules are executed in parallel. For every module in the architecture, the hidden layers are in a sequential order. The role of the components in the forecast architecture is to learn the features such as temporal dependencies and external factors of metro passen-

ger flow, thus obtaining a richer initial description of passenger flow with a restricted training set size. The base predictor in this forecast architecture is shown in Figure 4. This predictor can learn different passenger flow characteristics via different components. Therefore, the knowledge obtained from training the complete data set before bagging processing is still applicable to each base predictor.

The essential reason why the ensemble learning method has stronger forecast ability and generalization compared with a single predictor is that it introduces perturbation in the forecast process. Such perturbation includes perturbation to the sample space and perturbation to the feature space. The introduction of various perturbation factors can change the forecast logic of a single predictor, thereby making it more probable to learn neglected local or specific information [41]. Bagging perturbs the training data via bootstrap sampling, that is, perturbs the sample space. Therefore, the purpose of introducing bagging into the forecast architecture is to increase the perturbation in the forecast process to train diverse base predictors, thereby improving the final forecast accuracy. Since bagging changes the sample space of each base predictor, transfer learning can be successfully applied: bagging generates a lot of training subsets via bootstrap sampling, which are used to generate a lot of different base predictors. For every base predictor, its target domain D_T^i is different from each other and is a subdomain of the source domain D_S , but its target learning task T_T^i and source learning T_S are still the same. In addition, since the complete training set contains the data outside of each training subset, it can help each base predictor learn additional knowledge. Furthermore, transfer learning can accelerate the training speed of neural networks, so that the forecast architecture has better timeliness in practical applications. Transfer learning is achieved by using pre-training weights in the neural network. This section discusses the modules of the forecast architecture and the training algorithm in detail.

3.1 | External factors

Metro as an important part of urban comprehensive transportation, its passenger flow is closely related to local social

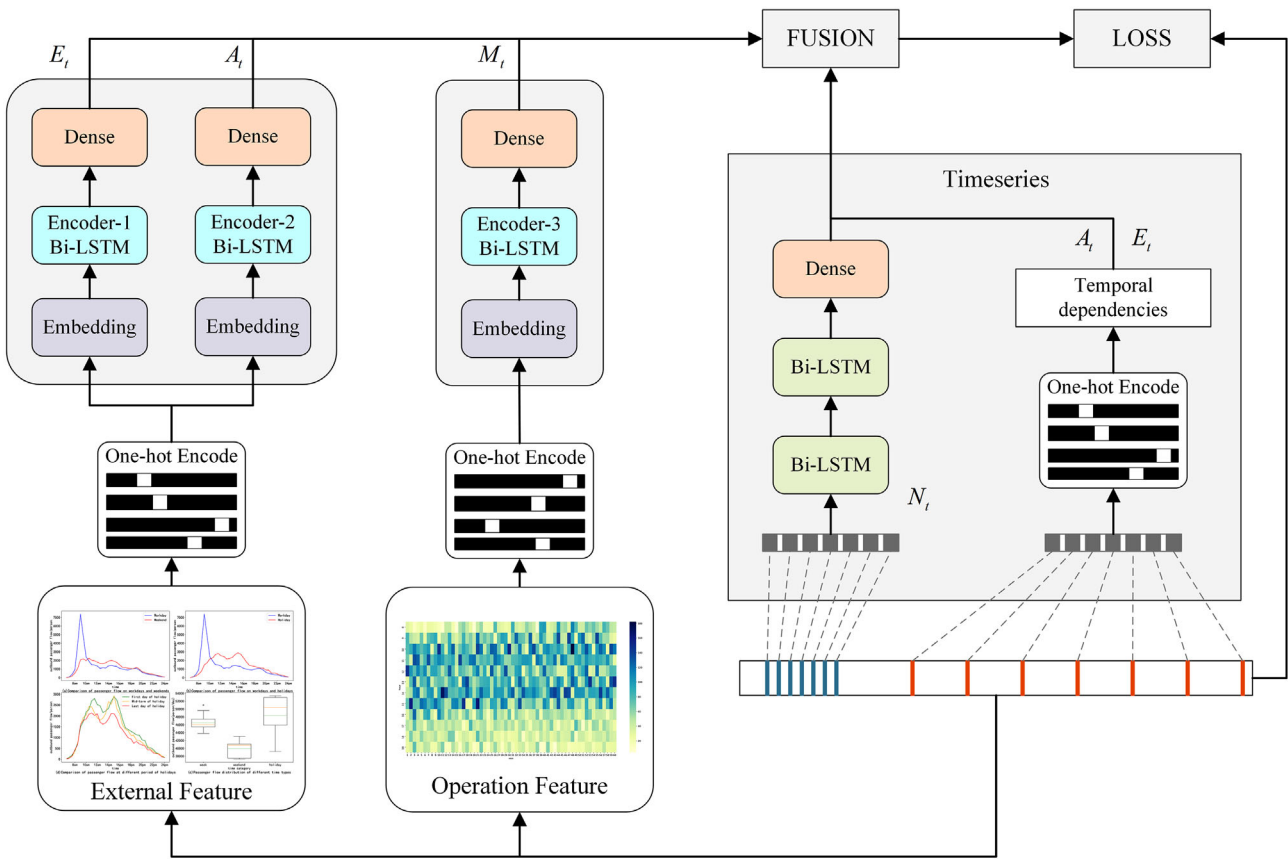


FIGURE 4 The base predictor in forecast hybrid architecture

and economic activities. Therefore, many related external factors will affect the accuracy of passenger flow forecast. This study will analyze and compare the specific impact of external factors on passenger flow, and propose factor learning methods. According to the changes in the work schedule of most travellers, every day can be divided into three types: workdays, weekends and holidays. The travel demand of travellers in different time types of a day will change. Take Chengdu Metro Tianfu Square Station as an example. Figures 5a and 5b show the passenger flow during a weekday, weekend and holiday, compared to a workday, holiday and weekend that have an evident influence on the outbound passenger flows at different times. For a holiday that includes multiple days, the passenger flow is different in different stages. As shown in Figure 5c, further analysis reveals that there are evident differences in daily passenger flow during the first day, mid-term and last day of holiday. Figure 6d (box diagram) shows the distribution of outbound passenger flow at a metro transfer station on all workdays, weekends and holidays in April 2018. As can be seen from the figure, the outbound passenger flow on weekends is evidently smaller than that on workdays and holidays, while the outbound passenger flow on holidays is widely distributed, generally more than on workdays and weekends. Through the above analysis and comparison, it can be seen that the total number of passenger flow in a day of different time types and the distribution of passen-

ger flow in different time periods are quite different. Therefore, taking the above-mentioned factors into consideration in the passenger flow forecast process may improve accuracy. Since the above-mentioned features are not continuous numerical variables but discrete categories, one hot encoding is used to encode the discrete features in this forecast architecture.

Define the outbound passenger flow data set $D_t = \{x_1, x_2, x_3, \dots, x_T\}$, where T is the number of outbound passenger flow data at a fixed time interval. For every $x_i \in D_t$, add a time type feature encoding that is encoded by a 10-dimensional one hot $E_i = [e_1, e_2, \dots, e_{10}]$, where e_1 to e_5 represent Monday to Friday of the workdays in sequence, e_6 and e_7 represent Saturday and Sunday of the weekends respectively, and e_8, e_9 and e_{10} represent the first day, mid-term and last day of holidays. For every $e_j \in E_i$, when $e_j = 1$ denote that the outbound passenger flow x_i belongs to j th type of time, otherwise $e_j = 0$. For every $x_i \in D_t$, add a time period feature encoding that is encoded by an 18-dimensional one hot $A_i = [a_6, a_7, \dots, a_{23}]$, where a_6 to a_{23} represent each hour from 6:00 to 23:00 during the metro operating period. For every $a_j \in A_i$, when $a_j = 1$ denote that the outbound passenger flow x_i at j am(pm), otherwise $a_j = 0$.

There are still some problems in the actual application of the above-mentioned encode method: firstly, each day is set as an individual dimension, but in fact, the passenger flow of the same time type may be similar. Secondly, when the number of

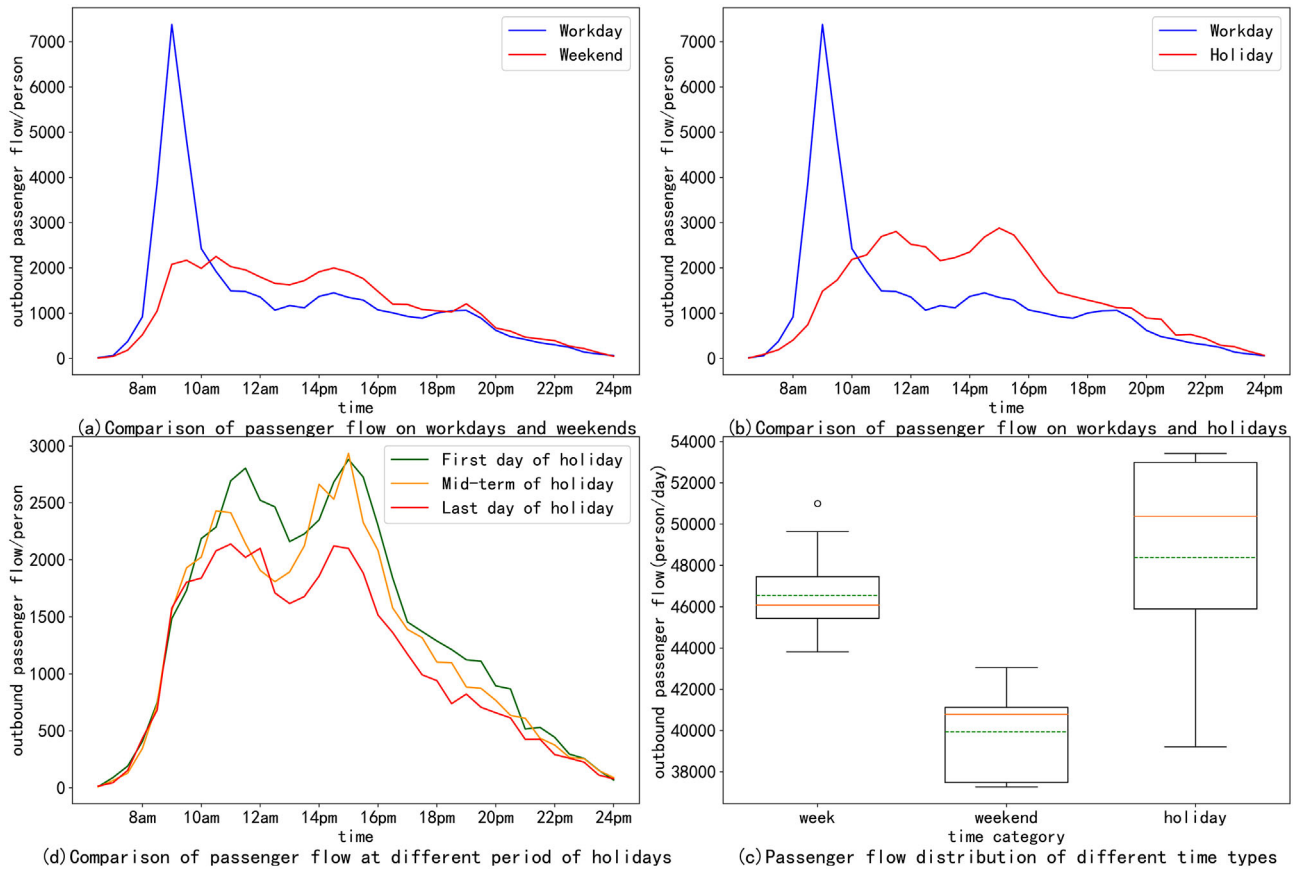


FIGURE 5 Comparison of passenger flow on workdays, weekends and holidays

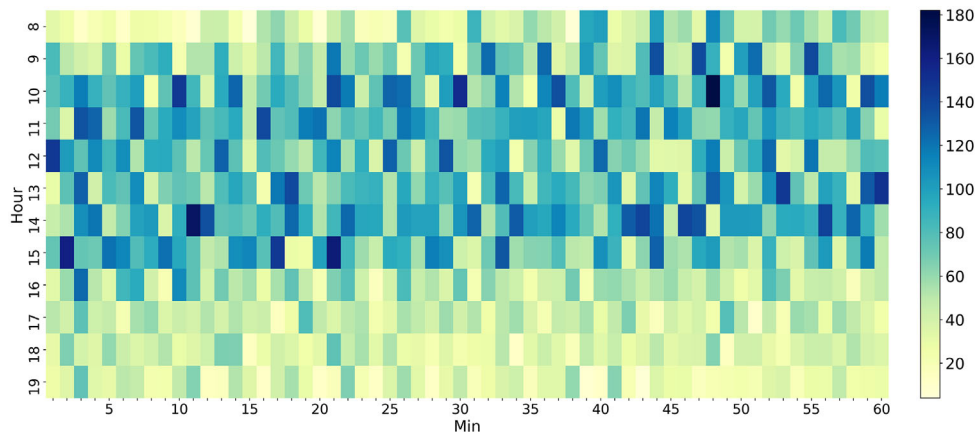


FIGURE 6 Heat plot for outbound passenger flow (1-min interval, 8:00–19:00)

categories is large, this is too sparse for one hot encoding, and the deep learning models do not work well with the highly sparse dimensional feature.

Embedding technique [42] is used to solve the above-discussed problems associated with one hot encoding. Embedding is widely used in the field of natural language processing [43]. It can map high-dimensional sparse vectors to dense low-dimensional space. Embedding can be represented as a mapping

$$Y = E(X) \quad (10)$$

where the function $E(\cdot)$ is injective and has a structure-preserving characteristic. Embedding in this architecture is different from that in word2vec which use unsupervised pre-training. It is a trainable component of the feature processing sub-model in the overall architecture. Therefore, embedding can be used to learn the low-dimensional and intensive effective

representation of the above-mentioned features in passenger flow. We use an Embedding layer and an encoder composed of Bi-LSTM to learn external factors and all layers are in sequential order, as shown in Figure 4.

3.2 | Temporal dependencies

When forecasting the outbound passenger flow x_t in the t th time interval, we usually use the outbound passenger flows in nearest time interval $N_t = [x_{t-b}, \dots, x_{t-2}, x_{t-1}]$, where b is the number of forecast timesteps. The relatively fixed travel patterns of travellers on workdays, weekends and holidays determine that passenger flow has a more significant periodicity than other time series. For instance, a metro passenger flow at a fixed time interval on Tuesday will be similar to a passenger flow at the same time interval on Monday. Considering periodic characteristics in the passenger flow forecast model can improve the accuracy of passenger flow forecast [19].

In this study, the above-mentioned time type feature E_i and time period feature A_i are combined to represent the periodicity of passenger flow. Specifically, E_i represents that passenger flow x_i belongs to a certain day of workdays, weekends or holidays, and A_i further represents which time period x_i belongs to that day. In the process of model training, when two feature encodings are input at the same time, the passenger flow of the same time type and in the same time period can be marked the same. Through sufficient learning on the training set, the neural network can learn the periodic characteristics of passenger flow autonomously, thereby improving the forecast accuracy of the model.

3.3 | Characteristics of metro operation

This study considers the characteristics of metro operation that are closely related to subway passenger flow. Figure 6 shows the outbound passenger flow of a metro station with a heat plot. The horizontal axis represents the 60 min in an hour while the vertical axis represents the metro operation time from 8:00 to 19:00. Each grid represents the outbound passenger flow aggregated at the 1-min interval for a whole day. A grid with deeper colour implies large outbound passenger flow. It can be seen from the figure that the outbound passenger flow of the metro is not evenly distributed within a fixed time interval, but takes the form of pulses. When the train arrives at the station, passengers will begin to disembark, resulting in a pulse of the outbound passenger flow. The operation of trains strictly abides by the train diagram, so the above-mentioned impulse passenger flow will appear in a regular pattern according to the train diagram. In the practical forecast, for the passenger flow x_i to be forecasted, the train diagram corresponding to the i th time interval is known. Therefore, we can get the pattern of pulse passenger flow in the i th time interval in advance. Using the above-mentioned impulse mode as one of the input features of the model probably improves the forecast accuracy. For every $x_i \in D_i$, add $M_i = [m_i^1, m_i^2, \dots, m_i^n]$ to represent the pulse pattern of the outbound passenger flow in the i th time interval,

where n denote that the i th time interval contains n -min. For every $m_i^j \in M_i$, where $m_i^j = 1$ denote that there is a train arriving at the station within the j th minute of the i th time interval, otherwise $m_i^j = 0$.

Due to the lack of train diagram, this study uses a data-driven method to extract train diagrams and timetable information corresponding to passenger flow. The process is as follows: for every $x_i \in D_i$, if the outbound passenger flow at the j th minute in the i th time interval $x_i^j > x^m$, where x^m is the median of all outbound passenger flows aggregated at 1-min interval, let $m_i^j = 1$ represent that there is a train arriving at the j th min. Otherwise, $m_i^j = 0$. Restricted by factors such as train control system and turnaround of train stock, trains usually have a minimum departing interval. This means that in practical metro operation, the arrival time of two adjacent trains at the same station will be restricted by the minimum departing interval. According to the above encoding method, when outbound passenger flow is congested, there are probably several adjacent minutes of outbound passenger flow greater than x^m . At this time, the above adjacent minutes will be marked as having trains arriving at the station, which obviously does not meet the minimum departing interval constraint. Therefore, in the above encoding process, a minimum departing interval constraint is added to prevent this kind of situation from happening: when $m_i^j = 1$, means that a train arriving at the station at the j th minute of the i th time interval, let

$$\sum_{u=1}^r m_i^{j+u} = 0 \quad (11)$$

where r is the number of minutes of minimum departing interval. We use an Embedding layer and another encoder composed of Bi-LSTM to learn characteristics of metro operation and all layers are in sequential order, as shown in Figure 4.

3.4 | Fusion module

This module aims to learn the complex non-linear relationship between the output of the above-mentioned feature module and the output of the time series module, and then obtain the forecast results of the multi-feature after fusion. For different application scenarios, we can choose a different combination of modules. The fusion network in this study is shown in Figure 7.

First, use the concatenate layer to concatenate the output of each module, and then input the concatenated output into a MLP which consists of dense layers stacked sequentially to learn the complex non-linearity between each feature and the observation value relation.

3.5 | Training algorithm combining bagging and transfer learning

The combined application of bagging and transfer learning is illustrated in Figure 8. We use metro AFC data to construct

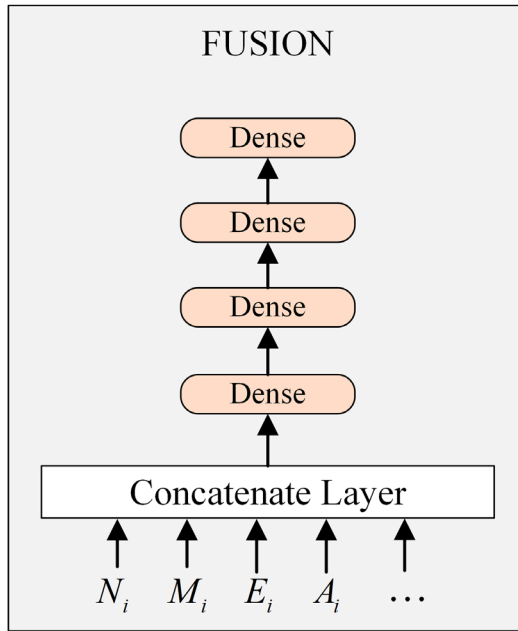


FIGURE 7 Fusion module

a dataset. Then the model is trained via backpropagation and Adaptive Moment Estimation (Adam) method [44].

The algorithm consists of four stages, as shown as follows:

Algorithm:

Stage I: Extract passenger flow features and obtain training set.

1. Input: Data from AFC
2. Step 1: Resample the input at a fixed time interval q to get the data set $D_t = \{x_1, x_2, x_3, \dots, x_T\}$
3. Step 2: Use input to add adjacent passenger flow sequence and passenger flow features to each timestamp passenger flow in D_t , including the following four steps.
 - (1) For every $x_i \in D_t$, add time series of passenger flow in nearest time interval $N_i = [x_{i-b}, \dots, x_{i-2}, x_{i-1}]$. The b is the number of forecast timestep.
 - (2) For every $x_i \in D_t$, encoding its external factors E_i, A_i . (Section 3.1)
 - (3) For every $x_i \in D_t$, encoding its temporal characteristics. (Section 3.2)
 - (4) For every $x_i \in D_t$, encoding its metro operational characteristics M_i . (Section 3.3)

4. Step 3: Sample $\{N_i, E_i, A_i, M_i, x_i\}$ make up the training set.

5. Output: The training set.

Stage II: Pre-training with base predictor.

6. Input: Training set.
7. Step 4: For every sample $\{N_i, E_i, A_i, M_i, x_i\}$ in training set, put N_i into the time series module, put E_i into the time type module, put A_i into the time period module, put M_i into the metro operational characteristics module, and x_i as the train label.
8. Step 5: Train multiple input and single output model through backpropagation and Adam algorithm.
9. Output: The neural network weight K_w of the model after the training is complete

Stage III: Combine bagging and transfer learning to train different base predictors

10. Input: Training set.
11. Step 6: Bootstrap sampling in the training set to get n subsets
12. Step 7: Initialize n base predictors and load the pre-training weight K_w into each base predictor.

13. Step 8: Input the n sub-datasets obtained in Step 6 into the n base predictors in Step 7 for training, and the training procedure is the same as Stage II.
 14. Output: n trained base predictors
- Stage IV: Fusion and prediction.
15. Input: (1) Testing set (2) n base predictors obtained in Stage II.
 16. Step 9: For every sample (N_j, E_j, A_j, M_j) in testing set, put N_j into the time series module of n base predictors respectively, put E_j into the time type module of n base predictors respectively, put A_j into the time period module of n base predictors respectively, put M_j into the operation module of n base predictors, respectively.
 17. Step 10: Use n base predictors to forecast separately and get n forecast results $\{\hat{x}_1, \hat{x}_2, \dots, \hat{x}_n\}$.
 18. Step 11: Calculate the median of n forecast results in Step 10 to get the final prediction result $\hat{x}$.
 19. Output: $\hat{x}$

4 | EXPERIMENTS AND RESULTS

4.1 | Dataset division and preprocessing

This study uses AFC data from Tianfu Square Station of Chengdu Metro Line 1 to verify the above-mentioned metro outbound passenger flow forecast architecture based on ensemble learning and deep learning, as shown in Table 1. Tianfu Square Station is one of the main transfer stations on Chengdu Metro Line 1, and it is also an important transportation hub of Chengdu's urban comprehensive transportation system. The data consist of all records during metro operate period 6:10 to 23:50 from 1 April to 30 April 2018 (including 20 workdays, 5 weekends and 5 holidays, a total of 1,373,014 samples). Abnormal data was removed before constructing the final dataset. With reference to related studies in [19] and [1], we choose a fixed time interval of 10 min for gathering passenger flow, and the daily passenger flow consists of 106 timestamps, from 6:10 to 23:50. Choose 22 April (weekend), 23 April (workday) and 29 April (holiday) as the test set, and the rest of the data is the training set (including 19 working days, 4 weekends and 4 holidays). The data on weekends and holidays in the training set accounted for a small proportion. Therefore, when bootstrap sampling and randomly dividing the validation set, it is very likely that the proportion of weekends and holidays data in the sub-data set and validation set will be further reduced, resulting in a decrease in forecast accuracy. In order to solve the above problems, this study maintained the proportion of workdays, weekends and holidays in the training set during bootstrap sampling and validation set division. It is ensured that the proportions of the above three time types of data in each bagging random subset and validation set are the same as the overall training set.

In order to speed up the learning and convergence when training our model [23], min-max normalization is performed to scale the data to the range of $[0, 1]$, as shown in Equation (12). Then we use a sliding window approach to sample the scaled data and reshape them into 3-D tensor with required form. In the evaluation, the predicted result is mapped back to the original data space and compared with the actual observation value.

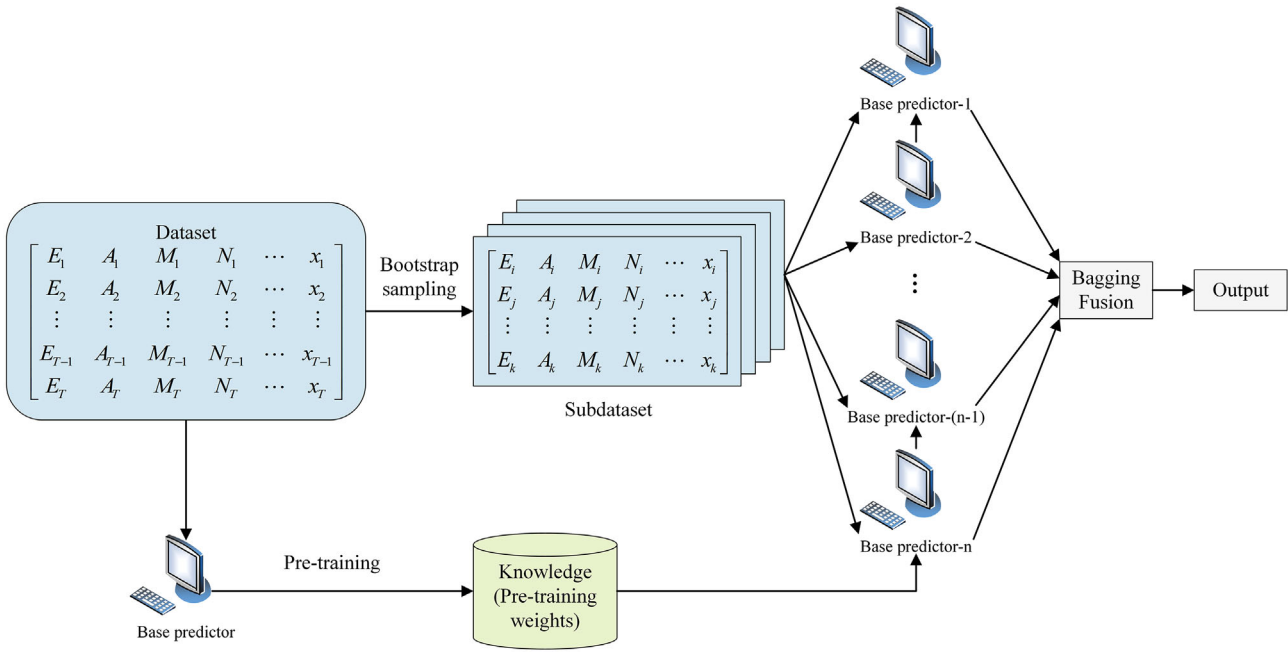


FIGURE 8 Combined application of bagging and transfer learning

TABLE 1 The AFC data of Chengdu Metro Tianfu Square station

ID	Inbound station code	Inbound time	Outbound station code	Outbound time
10114219	326	2018/04/08 17:08:09	126	2018/04/08 17:28:27
10303471	221	2018/04/08 16:54:09	126	2018/04/08 17:30:40
10556087	143	2018/04/08 16:52:03	126	2018/04/08 17:31:45
⋮	⋮	⋮	⋮	⋮

$$\tilde{x} = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (12)$$

4.2 | Evaluation metrics and experimental settings

Experiments were conducted to compare the prediction performance of the architecture proposed in this study with ARIMA [45], LSTM [4] and GRU [46, 47]. In addition, the prediction performance of the architecture proposed in this study was also compared with the hybrid prediction model DeepPF (DPF) proposed by Liu et al. [19]. Use grid search to automatically optimize the key parameters p (AR term), d (The order of difference) and q (MA term) of ARIMA. The LSTM model is composed of 2 LSTM layers and 1 Dense layer stacked sequentially. The number of hidden units is 256, 128 and 1, respectively, and the activation function is relu [48]. The GRU model is also composed of two GRU layers and one Dense layer stacked sequentially. The numbers of hidden units is 128,

64 and 1, respectively, and the activation function is relu. The constructed hybrid model DPF was adjusted according to the data set of this experiment and hyperparameters were adjusted. The study in [49] revealed that the evaluation performance of r^2 -score and symmetric mean absolute percent error (SMAPE) in supervised machine learning regression is better than Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE). Therefore, the performance metrics used in the study are Mean Absolute Error (MAE), SMAPE and r^2 -score as shown in Equations (13) to (15).

$$SMAPE = \frac{2}{N} \sum_{i=1}^N \frac{|x_i - \hat{x}_i|}{x_i + \hat{x}_i} \times 100\% \quad (13)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i| \quad (14)$$

$$r^2\text{-score} = 1 - \frac{\sum_{i=1}^n (\hat{x}_i - x_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (15)$$

TABLE 2 The hyperparameters of base predictor

Module	Layers(sequential)	Units	Activations
Time period	Embedding-BiLSTM-BiLSTM-Dense	(-,32,4,1)	(-,tanh,tanh,linear)
Time types	Embedding-BiLSTM-BiLSTM-Dense	(-,16,4,1)	(-,tanh,tanh,linear)
Metro operation	Embedding-BiLSTM-BiLSTM-Dense	(-,16,4,1)	(-,tanh,tanh,linear)
Time series	BiLSTM-BiLSTM-Dense	(256,128,1)	(relu,relu,linear)
Fusion	Dense-Dense-Dense-Dense	(500,200,50,1)	(relu,relu,relu,linear)

TABLE 3 Performance of different models in test data set

Model	SMAPE	The increase of performance (SMAPE)	MAE	The increase of performance (MAE)	r2_score
ARIMA	62.49%	-265.65%	242.41	-440.97%	0.0462
LSTM	17.09%	0	44.81	0	0.9698
GRU	14.61%	14.51%	40.03	10.67%	0.9776
DPF	13.65%	20.13%	38.61	13.84%	0.9772
D-E	18.73%	-9.60%	40.83	8.88%	0.9742
D-EA	15.84%	7.31%	40.17	10.35%	0.9779
D-EAM	14.25%	16.62%	34.07	23.97%	0.9834
DBT-EAM	13.60%	20.42%	33.03	26.29%	0.9841

where x_i is the observed passenger flow in the i th time interval, $\bar{x}_i$ is the average of passenger flow, $\hat{x}_i$ is the forecast value during i th time interval, N is the capacity of the test sample.

All experiments are based on Python3.8 and TensorFlow2.3 and computer with an NVIDIA RTX2060 and 16-GB memory. For LSTM and all base predictors, batchsize = 16, timesteps = 4 and epochs = 1000. Earlystopping is used in the training process, the optimizer is Adam and the loss function is MAE. The number of base predictors of the forecast architecture is 10, and the sample size of the random subset is 1800. Other hyperparameters of the base predictor model are shown in Table 2.

4.3 | Forecast performance comparison and analysis

We compare the forecast performance of the metro outbound passenger flow forecast architecture proposed in this study with the above models. In addition, we obtained different variant models by combining different passenger flow feature modules, and specifically analyzed how the above-mentioned passenger flow features affect the forecast accuracy.

D-E denotes the base predictor with time type features.

D-EA denotes the base predictor with time type features and time period features.

D-EAM denotes the base predictor with time type features, time period features and metro operational characteristics.

DBT-EAM denotes the forecast hybrid architecture with time type features, time period features and operational characteristics.

The forecast performance of each model on the test set is shown in Table 3. The performance of the LSTM is used as the standard to measure the increase of performance of other models. ARIMA has the worst forecast performance among all models, with r2_score only 0.0462. The performance of LSTM is significantly better than ARIMA, with r2_score of 0.9698. GRU performs the best among the traditional forecast models with r2_score of 0.9776, which is higher than the LSTM and D-E models. In terms of MAE and SMAPE metrics, the GRU model has significantly better forecast performance than LSTM, and even slightly better than two base predictors, D-E and D-EA. The hybrid model DPF performs well, with significantly better forecast performance than all traditional models and better than the D-EA model. Compared with the LSTM model, the base predictor gradually increases the forecast accuracy as passenger flow features are gradually added. The forecast performance of the D-EAM model with the introduction of operational characteristics is significantly better than that of the DPF model. In addition, the DBT-EAM model proposed in this study has achieved the best forecast performance in all evaluation metrics. From the results in Table 3, the performance of the ARIMA model seems to have a large gap compared to the benchmark model LSTM. Similar results were obtained by Yang et al. [1] and Siامي-Namini et al. [50] in their related studies in the fields of short-time passenger flow forecasting and financial time series forecasting. This is most likely due to the limitations

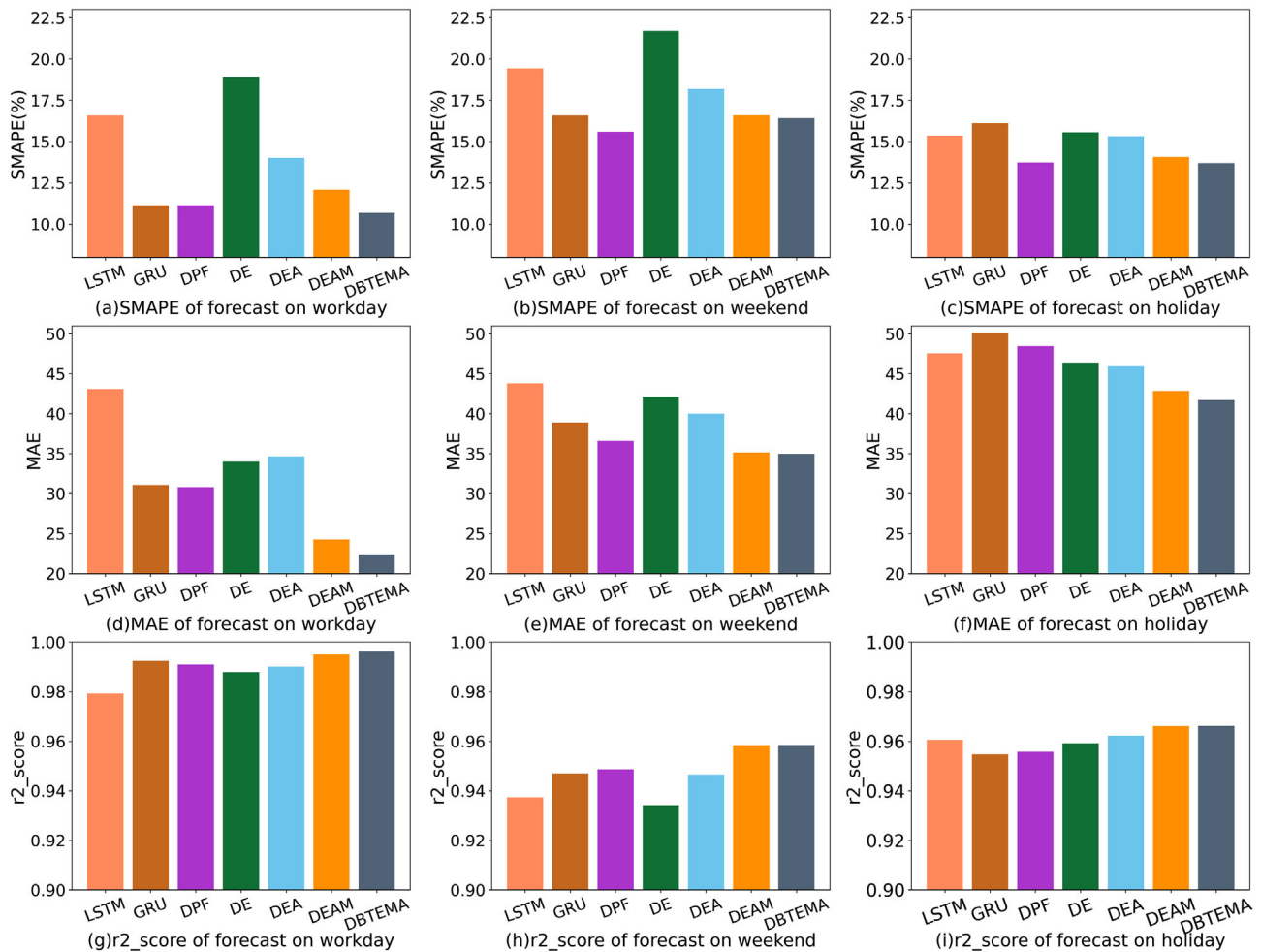


FIGURE 9 Forecast performance of different models in different time types

of the ARIMA model itself, as the traditional ARIMA model can essentially only capture the linear relationship of stationary time series or non-stationary time series that after differencing [50, 51]. However, the time series of short-time passenger flow can hardly meet the above requirements. It has more significant time-variance, randomness and strong non-linearity than medium- and long-term passenger flow, and is affected by many other factors such as time periods, holidays, workdays and even weather.

Through the above analysis, we know that the number of outbound passenger flow and passenger flow distribution on workdays, weekends and holidays are very different. Therefore, it is necessary to specifically analyze the forecast performance of each of the above models on different time types of test sets. Figure 9 and Table 4 exclude the worst-performing ARIMA model, and compare the forecast performance of the other seven models on workdays, weekends and holidays. As shown in Figure 9 and Table 4, the performance of the D-E model with time type features in the workdays is greatly improved compared with LSTM, and the MAE is reduced by 21.10%. However, on weekends and holidays, the performance of the D-E model is not significantly improved compared to LSTM. The D-EA

model with time type and time period features includes the external and temporal dependency features of passenger flow, so its forecast performance in all time types is better than the LSTM model and the D-E model.

Among all passenger flow features, the introduction of metro operation characteristics has the greatest improvement in the performance of the base predictor. Compared with LSTM, on workdays, weekends and holidays, the SMAPE of the D-EA model is, respectively, reduced by 15.44%, 6.43% and 0.39%. And the MAE is, respectively, reduced by 19.64%, 8.72% and 3.43%. The performance improvement of D-EAM, which adds metro operation characteristics to D-EA, is very significant. Compared with LSTM, on workdays, weekends and holidays, the SMAPE is, respectively, reduced by 27.08%, 14.61% and 8.52%. The MAE is, respectively, reduced by 43.71%, 19.82% and 9.91%. In addition, the experimental results revealed that the forecast performance of the D-EA model without metro operational characteristics is slightly worse than that of the GRU model and the DPF hybrid model, while the D-EAM model with operational characteristics has a significant advantage over the above two models. This reveals that metro operation characteristics are one of the important factors that affect the

TABLE 4 Performance of models in different time types

Type of time Model	Workday			Weekend			Holiday		
	SMAPE	MAE	r2_score	SMAPE	MAE	r2_score	SMAPE	MAE	r2_score
LSTM	16.58	43.08	0.9793	19.44	43.80	0.9373	15.37	47.54	0.9606
GRU	11.14	31.07	0.9923	16.58	38.88	0.9470	16.11	50.15	0.9548
DPF	11.63	30.81	0.9909	15.60	36.58	0.9487	13.72	48.44	0.9557
D-E	18.92	33.99	0.9878	21.71	42.12	0.9342	15.56	46.38	0.9592
D-EA	14.02	34.62	0.9900	18.19	39.98	0.9466	15.31	45.91	0.9623
D-EAM	12.09	24.25	0.9950	16.60	35.12	0.9584	14.06	42.83	0.9660
DBT-EAM	10.68	22.41	0.9961	16.42	34.97	0.9585	13.70	41.69	0.9661

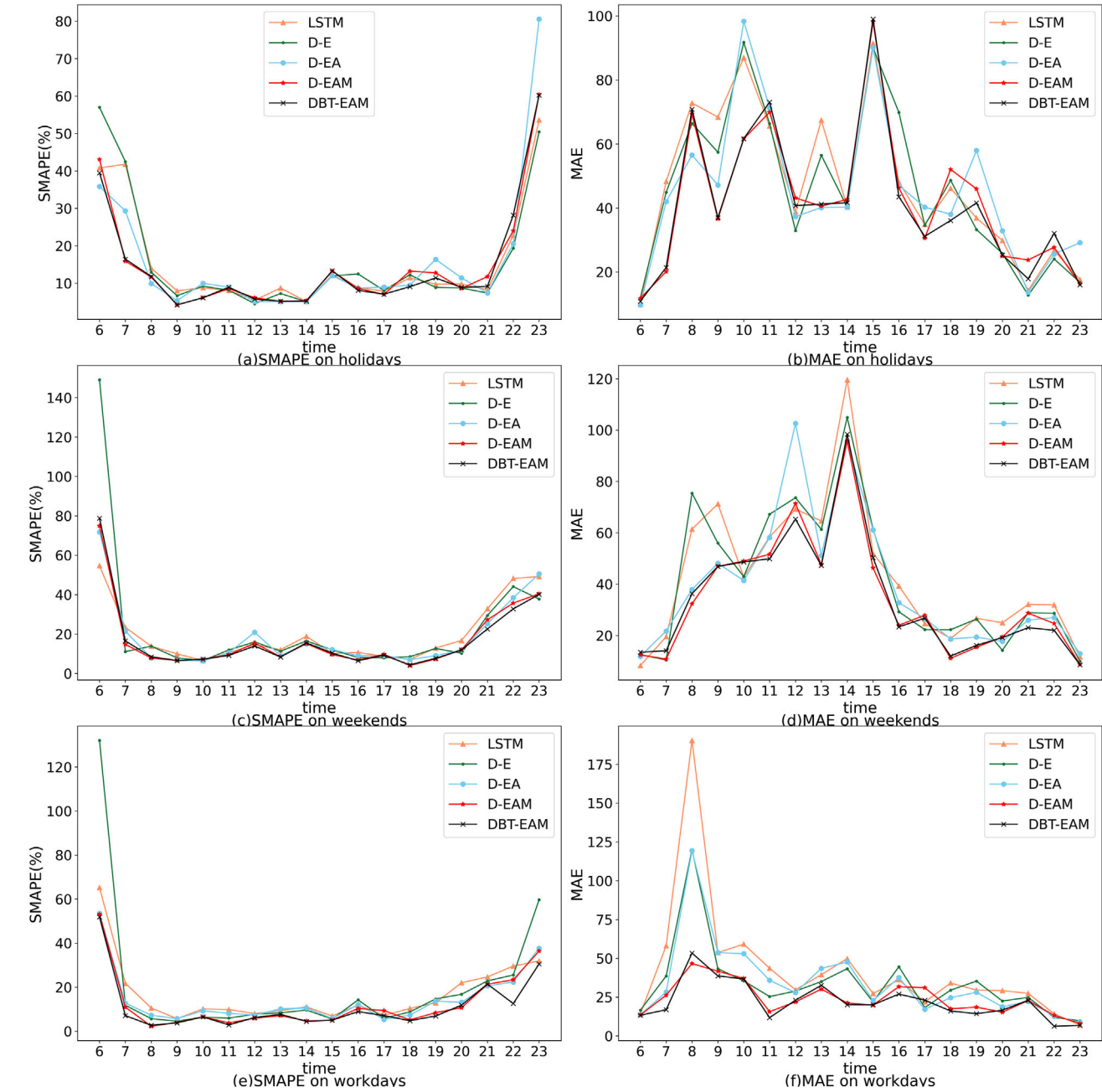


FIGURE 10 Performance of models during different times of the day

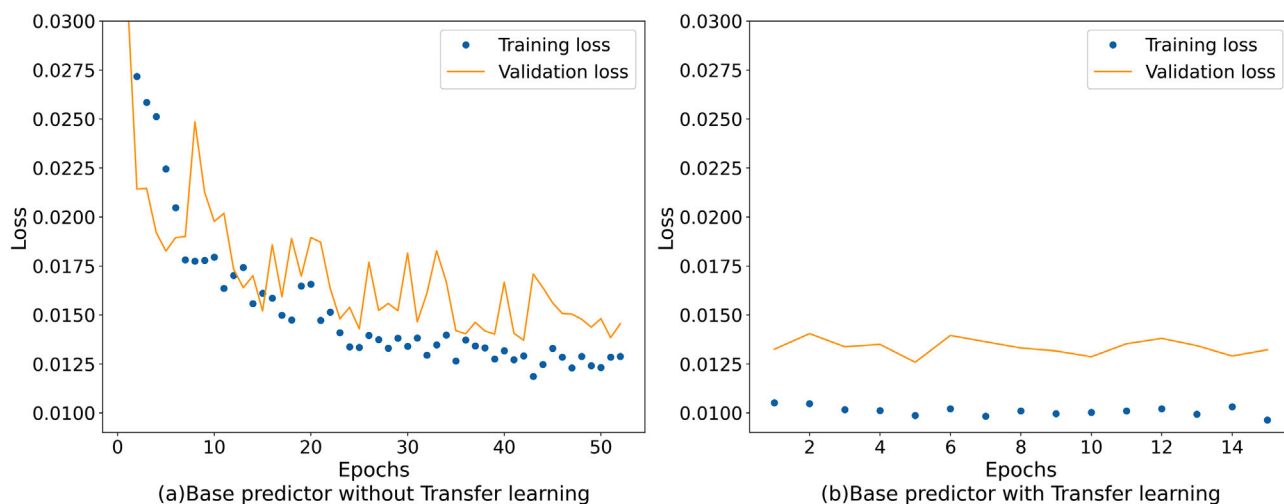


FIGURE 11 Comparison of the convergence of the training loss and validation loss of the two models

accuracy of passenger flow forecast, and the metro operation characteristics fusion method proposed in this study has excellent forecast performance. In addition, the DBT-EAM model, which integrates bagging and transfer learning and has all the above-mentioned passenger flow characteristics, has obtained the best forecast performance on workdays, weekends and holidays.

Figure 10 shows the SMAPE and MAE of benchmark model and the hybrid model of this study in each hour of the operating period during holidays, weekends and workdays. As shown in Figure 10, the forecast accuracy of the LSTM model at around 8 o'clock, 12 o'clock, 14 o'clock and 21 o'clock on holidays and weekends, and around 8 o'clock and 21 o'clock on workdays is significantly lower than that of other periods of the same day. From the distribution characteristics of passenger flow on holidays, weekends and workdays shown in Figures 5a and 5b, it can be seen that there has been a rapid increase or decrease in passenger flow during the above period. This indicates that the LSTM model has poor forecast ability during the sudden change period of passenger flow. And based on the base predictor proposed in this study and introducing different passenger flow characteristics, especially metro operation characteristics, can significantly improve the forecast accuracy during the sudden change period of passenger flow. Furthermore, the DBT-EAM model almost achieved the highest accuracy no matter during the sudden change period of passenger flow or other periods. The above results reveal that the DBT-EAM forecast hybrid architecture proposed in this study has excellent forecast performance and stability. Compared with the LSTM model, it can significantly improve the forecast ability during the sudden change period of passenger flow.

Figure 11 shows the convergence of training loss and validation loss when the base predictor with transfer learning and the base predictor without transfer learning are trained on the same random subset. Table 5 records the relevant data of the two models during the training process. Early stopping is used in the training process, and set patience = 10. As shown in Figure 11 and Table 5, the base predictor without transfer learning stops

TABLE 5 The relevant data of the two models during the training process

Model	Stopping epoch	Training time (s)	Minimal loss
Predictor without TL	52	175.95	0.0137
Predictor with TL	15	51.75	0.0126

after training for 52 epochs, the training time is 175.95 s, and the minimum validation loss is 0.0137. The convergence accuracy and convergence speed of the base predictor with transfer learning are significantly improved: it stops after training for 15 epochs, the training time is 51.75 s, and the minimum validation loss is 0.0126. The experimental results reveal that the fusion of transfer learning in the forecast architecture greatly improves the accuracy and the convergence speed of the base predictor. Based on the experimental platform of this study, the time to train a base predictor is reduced by about 70%. Since the forecast hybrid architecture proposed in this study is composed of multiple similar base predictors, the introduction of transfer learning can significantly reduce the training time of the architecture, while improving the forecast accuracy, increasing its timeliness and practicability.

5 | CONCLUSION

The contribution of the work is mainly on implementing a new hybrid architecture which is an end-to-end forecast hybrid architecture for metro outbound passenger flow that integrates multiple passenger flow features. The architecture integrates bagging and transfer learning with deep learning, and includes multiple extensible modules to learn the external factors, temporal dependencies and operational characteristics of passenger flow. In addition, this study proposes a new encoding and processing method for the operational characteristics of the metro. The forecast architecture is verified based on the AFC

data of Chengdu Metro Tianfu Square Station. The experimental results revealed that compared with the ARIMA, LSTM, GRU and hybrid model DPF, the forecast architecture proposed in this study has achieved the best forecast performance and forecast stability during workdays, weekends and holidays. Compared with the traditional LSTM model, this forecast architecture has the most significant performance improvement during the sudden change period of passenger flow. At the same time, the experimental results revealed that using the encoding and processing methods proposed in this paper to learn metro operational characteristics can significantly improve the accuracy of outbound passenger flow forecast. Furthermore, integrating transfer learning into the architecture not only improves the forecast accuracy, but also significantly accelerates the convergence speed of training, increasing the timeliness and practicality of the model.

Although the metro outbound passenger flow forecast architecture proposed in this study shows good performance, it still has some limitations and needs further improvement. First, although a more complex forecast model is constructed in this paper, its forecast performance is not significantly superior to that of the state-of-the-art model DPF. We hope to include as many passenger flow characteristics that affect the forecast accuracy as possible in the forecast process. Therefore, a certain degree of complexity in modelling is necessary. The modelling process and hyperparameters should be optimized more deeply for each sub-model dealing with passenger flow characteristics to maximize the forecast performance of the architecture, but this is not discussed further in this paper due to the limitation of the paper length. And this will be the emphasis of our future study. Second, the architecture can add more passenger flow feature processing modules, such as spatial features, weather features, to further improve the forecast accuracy. Third, due to the compute capability limitation of the platform, this study did not discuss the influence of the number of bagging-based predictors on the forecast accuracy. Finally, this study forecast the outbound passenger flow of a single station, and the forecast architecture can be further improved to forecast multiple stations at the same time.

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CONFLICT OF INTEREST

All authors disclosed no relevant relationships.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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